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Success Score: A Deep Learning Framework for Predicting Football Match Outcomes and Evaluating Team Performance

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ABSTRACT

This study presents a novel predictive framework for estimating football match outcomes and assessing team performance through a new metric called the Success Score. This metric integrates Expected Goals (xG) with actual scoring results to provide a comprehensive evaluation of both opportunity creation and offensive execution. A Deep Neural Network (DNN) was trained on three seasons (2020–2021, 2021–2022, 2022–2023) of match data from four major European leagues, incorporating features such as tactical play styles, rolling performance averages, and team quality indicators. The model demonstrated strong predictive performance, achieving a Mean Absolute Error (MAE) of 0.3142 and an R^2 score of 0.8592 in cross-validation. It also generalized effectively to out-of-sample matches from the 2024–2025 season. Furthermore, the Success Score allowed for the classification of outcomes, enabled outcome prediction with 73.30% accuracy for Win vs. Not Win and 75.13% for Lose vs. Not Lose. Beyond predictive accuracy, the framework offers interpretable insights into team dynamics, revealing patterns of overperformance and underperformance. Case studies of FC Barcelona and Manchester United illustrate their practical utility for tactical analysis and strategic planning. This scalable, data-driven approach advances modern football analytics by supporting continuous performance monitoring and informed decision-making.

INTRODUCTION

Predicting match outcomes in football remains a challenging task for coaches, analysts, and fans. This complexity arises from the interaction of various tactical styles, team dynamics, and situational factors, such as player form. Additionally, football differs from other team sports like basketball or ice hockey due to its low frequency of scoring attempts and goals, which amplifies its inherent unpredictability [1]. The fascination of football largely stems from its inherently complex nature. A multitude of factors, both on and off the field, interact to influence match outcomes, showcasing the sport's intricate and dynamic character. Off-field factors, such as travel fatigue, home-crowd advantage, and players' psychological states, significantly influence match outcomes. Buchheit et al. analysed over 61,000 football matches and demonstrated that travel distances influence match outcomes, underscoring the importance of off-field dynamics [2]. On-field factors, such as team formations, player quality, and tactical decisions, also play a major role in determining the result of a game. Yeung et al. emphasized the significance of team formations and FIFA player ratings in predicting match outcomes, highlighting the importance of tactical choices and player capabilities [3]. FIFA provides comprehensive data about football teams and player ratings. Nouraie et al. utilized the FIFA (SoFifa) dataset, also used in our study, to develop an intelligent player selection system. By leveraging neural networks and the Hungarian algorithm, they optimized starting line-ups based on player ratings, demonstrating the potential of machine learning in refining team formations and strategic decisions [4]. Nouraie and Eslahchi further contributed to this area by identifying key attributes that influence a player's success in specific positions. Their study offers a data-driven approach to assist teams in optimizing player selection and positioning, thereby enhancing overall team performance [5]. In football statistics and data science, team performance is evaluated using a wide range of parameters collected through advanced technologies such as computer vision and manual monitoring. Platforms like FBref provide extensive match data, including metrics such as passes, pass accuracy, long

balls, and shots. From this data, we have categorized tactical play styles by grouping related statistics into coherent patterns. In our analysis, we adopted a framework of ten primary football tactics, each representing a distinct strategic approach. Understanding these tactical profiles is essential in football analysis, as they shape how teams approach matches, leverage their strengths, and respond to their opponents. Strategic adaptability, whether to exploit a weakness or maintain control, often defines the outcome of a game. For example, Plakias et al. conducted a systematic review to identify soccer players' playing styles, focusing on their positions within team formations [6]. Building on this understanding of tactics and individual contributions, we present a structured summary of common tactical play styles and their associated measurable features. Table 1 outlines ten distinct tactical styles, each capturing a unique strategic approach in football. These styles can be quantified using specific performance metrics extracted from match data. For example, Possession Play emphasizes maintaining control through short passes and patient build-up, while Counter-Attack relies on quick transitions and exploiting space with long balls. High Press is characterized by intense pressure in advanced areas to regain possession, whereas Direct Play bypasses the midfield through rapid, vertical passing. By associating tactical styles with measurable indicators, we can incorporate tactical dimensions into predictive models, improving both their accuracy and real-world applicability.

Recently, metrics like Expected Goals (XG) have been introduced to offer a statistical measure of the quality of scoring opportunities, enabling a deeper understanding of team and player performance beyond final results [20]. Such measures have proven instrumental in bridging the gap between raw match outcomes and the underlying dynamics of gameplay. However, while XG provides valuable insights into scoring probabilities, it does not fully account for the influence of tactical styles and team strategies. Studies such as those by Fernandez-Navarro et al. [21] have highlighted the importance of including contextual and tactical variables such as pressing intensity, counter-attacking tendencies, and positional play into performance evaluations.

With the advent of machine learning, predictive modelling in football has experienced a significant transformation. Researchers have explored a variety of algorithms and frame- works to improve the accuracy of prediction. Ren and Susnjak [22] applied the Kelly index to classify matches based on predictive difficulty, benchmarking machine learning models against bookmaker odds. Constantinou [23] introduced the Dolores framework, a hybrid Bayesian network that integrates dynamic ratings to predict outcomes across multiple leagues. Mills et al. [24] employed logistic regression, XGBoost, and neural networks to predict results in multiple leagues, emphasizing data augmentation and feature selection. Huba´ek et al. applied gradient-boosted trees to predict soccer match outcomes using relational data, demonstrating improved predictive accuracy compared to traditional models [25]. Ievoli and Palazzo [26] demonstrated that passing network indicators that quantify player interactions improved forecasting models. These studies highlight the growing role of machine learning in predicting football outcomes and its potential to outperform traditional statistical methods.

Building on these advances, our study focuses on integrating tactical data with outcome prediction by introducing a new scalable statistical parameter, the Success Score. This continuous metric estimates a team's probability of success against a specific opponent by combining Expected Goals with Actual Goals Scored. By capturing both the quality and the realization of scoring chances, the Success Score offers a more nuanced assessment of team performance. Using match data from 2021 onward, we incorporate this metric into a modeling pipeline that reflects the evolving dynamics of modern football. This approach transforms traditionally qualitative aspects, such as team strength and tactical efficiency, into quantifiable predictive features.

Table 1. Tactical play styles and their associated metrics.
Summary of the ten tactical play styles used in this study and the match statistics used to compute each style score (metrics sourced from FBref). Abbreviations: xG, expected goals. References informing definitions: [7–19].

Tactical Play Styles	Identified Features from Data	References
Possession Play	Ball possession, total passes, pass accuracy (%), own half passes, XG from open play	[7, 8]
Counter-Attack	Low ball possession, big chances created, ac- curate long balls, opposition half passes, off- sides, shots inside the box	[9]
High Press	Fouls committed, opposition half passes, shots inside the box, corners, advanced throw-ins	[10]
Direct Play	Accurate long balls, shots inside the box, big chances created, offsides	[11]
Wing Play	Accurate crosses, shots inside the box, corners, advanced throw-ins	[12, 13]
Set Piece Focus	XG from set plays, corners, accurate crosses	[14]
Creative Playmaking	Pass accuracy (%), big chances created, ac- curate long	[15]

	balls, opposition half passes	
Playing Out from the Back	Accurate short passes, accurate long balls, build-up play from defensive third	[16, 17]
Low Block	Blocked shots, own half entries, fouls com- mitted, shots conceded from outside the box	[18]
High Defensive Line	Offsides forced, opposition half passes, fouls committed, shots inside the box conceded	[19]

METHODS

In this study, we develop a predictive model to estimate the success score, a novel performance metric designed to quantify and differentiate the performance of two competing football teams before a match occurs. This model leverages machine learning techniques and incorporates various tactical, statistical, and contextual features to generate pre-match predictions.

A. DATA SETS

To support this research, we utilized two key datasets: FBref Match Data and SoFIFA Team Attributes, which together provided a rich foundation for feature engineering and model development. We used data from 97 teams across four leagues: La Liga (Spain), Premier League (England), Bundesliga (Germany), and Serie A (Italy) for three seasons (2020–2021, 2021–2022, 2022–2023)

The FBref dataset contains match-level statistics from the top four European football leagues during three seasons (2020–2021, 2021–2022, 2022–2023). Key attributes include: Match Details: Team names, final scores, and (XG). Performance Metrics: Shots, ball possession, fouls, corners, and passes, among others.
This dataset served as the primary source for calculating the Success Score and other match-specific metrics, forming the core inputs for the predictive model.

The SoFIFA dataset provides detailed team-level attributes, including: Overall and Positional Ratings: Ratings for attack, midfield, and defence. Prestige Indicators: Measures of domestic and international prominence.

These attributes added contextual information about team quality and were instrumental in feature engineering.

B. SUCCESS SCORE DEFFENITION

The Success Score is a performance metric designed to evaluate a football team's effectiveness in a match. By combining actual goals scored with Expected Goals (XG)- which estimate the probability of a shot resulting in a goal based on various features of the shot- the Success Score provides a more comprehensive assessment of performance. This dual approach reflects both outcomes (goal-scoring efficiency) and opportunities (quality of chances created). The Success Score is calculated in three main steps:

I. Offensive Performance Metric: For each team, an offensive performance score is calculated as:

$$M(H) = G(H) + XG(H) + \beta \qquad (1)$$

$$M(A) = G(A) + XG(A) + \beta \qquad (2)$$

Where:
$G(H)$ and $G(A)$ denote the actual goals scored by the home and away teams.
$XG(H)$ and $XG(A)$ denote the expected Goals for the home and away teams.
$\beta = 0.5$ is a base value added to ensure minimum contributions for all teams.

II. Sigmoid Scaling: The offensive metrics are then scaled using a sigmoid function to avoid extreme values:

$$S(H) = \frac{1}{1+e^{-k(M(H)-x_0)}} \qquad (3)$$

$$S(A) = \frac{1}{1+e^{-k(M(A)-x_0)}} \quad (4)$$

Where:

k is sigmoid steepness. Sigmoid steepness controls how smoothly values transition. A lower k value (e.g., $k = 0.8$) results in smoother transitions.

$x_0 = 2$ is the transition point chosen based on the expected range of offensive metrics.

The sigmoid function addresses two key challenges:

Amplifying Small Values: Ensures meaningful contributions, particularly for teams with low goals or *XG*.

Flattening Large Values: Avoids extreme dominance by scaling down outliers.

Additionally, the inclusion of a base parameter $\beta = 0.5$ in the offensive performance metric ensures fairness by preventing negligible contributions from teams with zero goals or *XG*

III. Success Score Calculation: The final Success Score is calculated as an interdependent metric:

$$SS(H) = \frac{S(H)}{S(H)+S(A)} \times 10 \quad (5)$$

$$SS(A) = 10 - SS(H) \quad (6)$$

This interdependent calculation measures not only offensive strength but also includes defensive contributions, as stronger defences reduce the opposing team's Success Score.

FEATURE ENGINEERING

While the actual Success Score can be calculated after a match based on a team's performance in terms of Expected Goals and goals scored, its real value lies in being able to estimate it before a match is played. A reliable pre-match prediction enables teams, analysts, and decision-makers to assess the likely performance of a team against a specific opponent, aiding in tactical planning, player selection, and match preparation. To make this possible, we engineered a comprehensive set of features that encapsulate team dynamics and match context. These features were derived from tactical play styles, rolling averages of recent performance metrics, and static team attributes. By leveraging both contextual and statistical information, the model aims to provide a robust and interpretable forecast of team performance.

A. Tactical Play Style Scores

The Tactical Play Style Score quantifies a team's performance based on various tactical approaches during a match. Each score is calculated using the average of relevant match metrics associated with a specific tactical play style as outlined in **Error! Reference source not found.**. The general formula for calculating the Tactical Play Style Score (T_{style}) is:

$$T_{\text{style}} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad (7)$$

Where:

T_{style} is the score for a specific tactical play style (e.g., Possession Play, Counter Attack, etc.).

X_i represents the value of the i-th metric associated with that tactical play style.

n is the total number of metrics considered for the given play style.

For each match, the Tactical Play Style Scores are calculated separately for the home and away teams. These scores are represented as vectors:

$$T_{\text{HOME}} = [T_1 \;\; T_2 \;\; T_3 \;\; T_4 \;\; T_5 \;\; T_6 \;\; T_7 \;\; T_8 \;\; T_9 \;\; T_{10}]$$

$$T_{\text{AWAY}} = [T_1' \;\; T_2' \;\; T_3' \;\; T_4' \;\; T_5' \;\; T_6' \;\; T_7' \;\; T_8' \;\; T_9' \;\; T_{10}']$$

Where:

T_1 and T_1' denote the Possession Play Score. T_2 and T_2' denote the Counter-Attack Score. T_3 and T_3' denote the High Press

Score. T_4 and T_4' denote the Direct Play Score. T_5 and T_5' denote the Wing Play Score. T_6 and T_6' denote the Set Piece Focus Score.T_7 and T_7' denote the Creative Playmaking Score.T_8 and T_8' denote the Playing Out from the Back Score. T_9 and T_9' denote the Low Block Score. T_{10} and T_{10}' denote the High Defensive Line Score.

B. Rolling Averages

To capture both short-term fluctuations and long-term trends in team performance, we calculate rolling averages for 14 metrics derived from Tactical Play Style Scores and Performance Outcome Metrics. These metrics include: Tactical Play Style Scores: 10 scores corresponding to various tactical approaches (e.g., Possession Play, Counter-Attacks, High Press, etc.). Performance Outcome Metrics: Success Scores, Goals Scored, Goals Conceded, and Points Earned.

Rolling averages allow the model to analyse both recent form and overall season consistency, providing a dynamic view of team performance. Short-term rolling averages capture fluctuations in performance over the last five matches, while long-term averages reflect sustained trends over the season. This dual approach ensures that the model accounts for evolving team dynamics and contextual factors, offering a nuanced understanding of team readiness and potential performance in upcoming games. The rolling averages for each team are calculated starting from the second game of the season, as at least one match is required to compute these metrics. For teams that have not yet played five matches, the rolling averages are computed using the available number of matches until the required window size is reached. The general formulas for calculating rolling averages are:

$$\mathrm{M}_{\mathrm{t}}^{\mathrm{short}} = \frac{1}{5}\sum_{\mathrm{i=t-4}}^{\mathrm{t}} \mathrm{M}_{\mathrm{i}} \quad , \quad \mathrm{M}_{\mathrm{t}}^{\mathrm{long}} = \frac{1}{\mathrm{t}-1}\sum_{\mathrm{i=2}}^{\mathrm{t}} \mathrm{M}_{\mathrm{i}} \qquad (8)$$

Where:
t is the current match week; M_i is a performance metric and:
M_t^{short} shows the short-term rolling average over the last five matches.
M_t^{long} shows the long-term rolling average over all previous matches in the season.
M_i shows the value of a performance metric in the i-th match.
t shows the current match week.

C. Dynamic Feature Vectors

For each match, we construct four feature vectors short-term and long-term vectors for both the home and away teams. These vectors encapsulate the 14 rolling average metrics for each team. The home and away feature vectors are structured as follows:

$$\left[ST_{\{home\}}, \quad LT_{\{home\}}, \quad ST_{\{away\}}, \quad LT_{\{away\}}\right]$$

Each vector includes:
10 rolling averages of Tactical Play Style Scores. 4 rolling averages of Performance Outcome Metrics (Success Scores, Goals Scored, Goals Conceded, and Points Earned).
Rolling averages offer several key advantages: Smoothing Variances: They reduce short-term variability and emphasize consistent trends in team performance. Dynamic Insights: By combining both short-term and long-term trends, rolling averages provide a dynamic understanding of team strengths and weaknesses across different phases of the season. Studies by Oliva-Lozano et al. (2020) [27] demonstrated the effectiveness of rolling averages in identifying worst-case scenarios in professional soccer, while Griffin et al. (2021) [28] highlighted their advantages over exponentially weighted moving averages for performance analysis. By integrating these metrics dynamically, our model captures the evolving nature of team performance and contextualizes tactical effectiveness and stability over time.

D. Team Quality and Prestige Features

Using the SoFIFA dataset, we added features reflecting team quality and reputation:
Quality Features (QF): Overall, attacking, midfield, and defensive ratings (on a 100-point scale). Prestige Features (PF): Domestic and international prestige scores (on a 10-point scale).

E. Input Vector Construction

For each match, we constructed 68-dimensional feature vectors by combining the engineered features for both the home and away teams:

$$V = \left[ST_{home}, LT_{home}, ST_{away}, LT_{away}, QF_{home}, QF_{away}, PF_{home}, PF_{away}\right]$$

These vectors formed the input to our predictive model.

Model Development

We employed a Deep Neural Network (DNN) with the following structure: Input Layer: 68 features. Hidden Layers: Four fully connected layers (256, 128, 64, 1 neurons) with Tanh activations and Dropout (0.2) regularization. Output Layer: A single neuron with linear activation, predicting the home team's Success Score.

A. Training Strategy

The model was trained using five-fold cross-validation. Within each training fold, we applied random oversampling to balance the data; validation and test splits were left untouched.

B. Hyperparameter Selection

The selection of hyperparameters was conducted iteratively, focusing on achieving optimal performance and robust generalization. The following hyperparameters were tuned through experimentation: Learning Rate: A learning rate of 0.001 was chosen for the Adam optimizer. Preliminary trials showed that this value achieved smooth convergence without oscillations, balancing training stability and convergence speed. Epochs: The model was trained for 40 epochs. This value was determined as optimal after testing a range of 20 to 50 epochs, ensuring stable convergence while avoiding overfitting or underfitting. Batch Size: A batch size of 64 was selected after evaluating options of 16, 32, and

Data augmentation and cross-validation. To address class imbalance while avoiding data leakage, random oversampling was applied only to the training split inside each cross-validation fold. Validation and test sets were never oversampled. Five-fold CV was used, with 80% of the data for training after within-fold oversampling and 20% held out for validation in each fold.

This hyperparameter configuration was iteratively refined, achieving a balance between predictive accuracy and model robustness. The final setup allowed the model to generalize effectively across varying football match scenarios. Error! Reference source not found. provides an overview of all steps of the method.

Results

This section presents a detailed evaluation of the model's predictive accuracy, generalization capabilities, and potential for real-world application. The analysis begins with an investigation into home and away success scores, followed by an evaluation of the model's performance on both test data and unseen data from the 2024–2025 season. Finally, success score ranges are defined and optimized for classifying match outcomes, with their application demonstrated through predictions of match results and team-level insights. As a case study, the performances of Manchester United and Barcelona in different leagues are analysed by comparing their predicted and actual success scores, providing a detailed discussion on their respective outcomes.

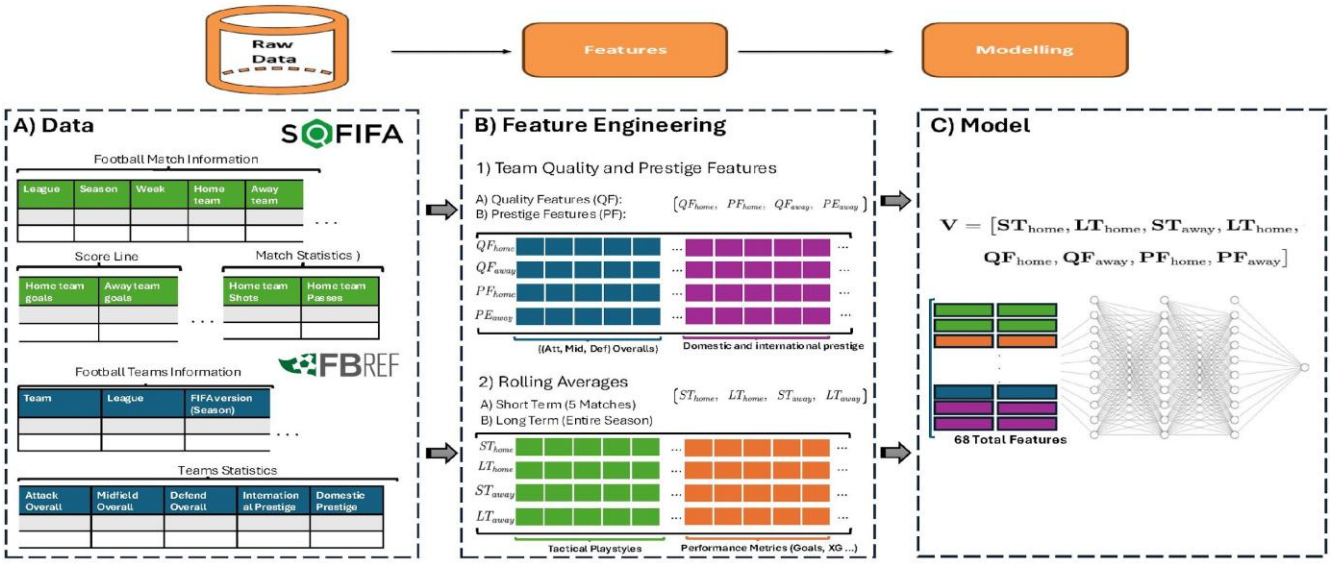

Figure 1. Overview of the modelling pipeline.

(A) Data acquisition from SoFIFA (team attributes) and FBref (match statistics). (B) Pre-processing and feature engineering, including computation of tactical play-style scores, short- and long-term rolling averages, and team quality/prestige features. (C) Deep neural network (DNN) training to predict the home Success Score, SS(H), on a 0–10 scale. Abbreviations: xG, expected goals; DNN, deep neural network; SS(H), home Success Score; SS(A), away Success Score.

A. Statistical Analysis of Home Success Scores Across Seasons and Leagues

To quantify home-ground advantage, we performed a one-sided paired t-test comparing team Success Scores at home versus away. The null and alternative hypotheses were H_0: μ_home ≤ μ_away; H_1: μ_home > μ_away. The test yielded t = 13.24 and $p = 3.18\times10^{-39}$, rejecting H_0 and supporting a home-ground advantage.

Building on this, we analysed how home success scores varied across leagues and seasons. The mean home success scores for each league and season are summarized in **Error! Reference source not found.**. The results reveal distinct league-level dynamics. La Liga teams exhibited the highest mean success scores across all three seasons, demonstrating a particularly strong home-ground advantage. By contrast, the Premier League and Serie A showed lower mean scores in certain seasons, reflecting more balanced competition or potentially weaker home dominance. These variations highlight the role of league-specific factors, such as fan influence, travel distances, and tactical preferences, in shaping home-ground performance. Overall, the findings underscore the measurable and persistent impact of home-ground advantage, while also highlighting how its magnitude can differ across leagues and evolve.

Table 2. Mean home Success Score by league and season.
Average home-team Success Scores (SS(H); 0–10 scale) across the English Premier League (ENG), La Liga (ESP), Bundesliga (GER) and Serie A (ITA) for the 2020–2021, 2021–2022 and 2022–2023 seasons. Higher values indicate stronger home performance. SS(H), home Success Score.

Season	ENG-Premier League	ESP-La Liga	GER-Bundesliga	ITA-Serie A
2021	5.0482	5.2094	5.1879	5.1291
2223	5.3227	5.4103	5.3451	5.2460
2324	5.2914	5.3613	5.3255	5.2929

B. Evaluation of Model performance

The model's performance was evaluated using a dedicated test dataset composed of matches not seen during training. This dataset included fixtures from the 2020–2021, 2021–2022, and 2022–2023 seasons, drawn from four major European football leagues: the English Premier League, La Liga, Bundesliga, and Serie A. The original dataset consisted of 4,213 match records. To address the limited dataset size and improve the model's ability to generalize, we applied random oversampling as a data augmentation strategy. This process expanded the dataset tenfold—from 4,213 to 42,130 samples—by duplicating existing records. Unlike synthetic augmentation techniques, this method preserved the original feature-target distribution and maintained the statistical integrity of the dataset. Random sampling ensured that all samples had an equal chance of being selected, minimizing the risk of introducing bias while enhancing pattern learning during training.
Following augmentation, a 5-fold cross-validation strategy was used to rigorously evaluate the model's performance. In each fold, 80% of the data (33,704 samples) was used for training, and the remaining 20% (8,426 samples) was reserved for testing. This procedure ensured that each sample contributed to both training and validation exactly once, enabling a comprehensive assessment of model stability. **Supplementary Figure S2** presents the training and validation loss curves across the five folds. The steady decline in training loss, accompanied by consistently low and stable validation loss, indicates effective learning and strong generalization capability.
To quantify model accuracy, we used three standard regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and the coefficient of determination (R^2). MAE provides a straightforward measure of average prediction error, MSE emphasizes the impact of larger errors, and R^2 reflects how well the model explains the variability of the target variable.
The results, summarized in **Error! Reference source not found.**, demonstrate the model's robust predictive performance. The average MAE across all folds was 0.3142 ± 0.0041, indicating low prediction error. The MSE averaged 0.2470 ± 0.0058, and the R^2 score reached 0.8592 ± 0.0045, highlighting the model's ability to capture and explain underlying patterns in team performance. Collectively, these results confirm the model's high accuracy, stability, and capacity to generalize effectively to

unseen match scenarios.

Table 3. Cross-validation performance on the training data.
Five-fold cross-validation results for predicting SS(H): Mean Absolute Error (MAE), Mean Squared Error (MSE) and coefficient of determination (R^2). Values are mean ± s.d. across folds. SS(H), home Success Score.

Metric	Result (Average ± Std)
MAE	0.3142 ± 0.0041
MSE	0.2470 ± 0.0058
$R2$	0.8592 ± 0.0045

C. Evaluation on Unseen Data (2024–2025 Season)

The previous section evaluated the model's performance during training and validation phases using historical data from the 2020–2021, 2021–2022, and 2022–2023 seasons. In this section, we assess the model's ability to predict the home Success Score ($SS(H)$) for matches in the 2024–2025 football season. This evaluation focuses on the model's accuracy in estimating $SS(H)$ values, providing insights into its ability to generalize beyond the temporal scope of its training data.
The model was trained on match data from four major leagues—English Premier League, La Liga, Bundesliga, and Serie A—and tested on completely unseen data from the first 11 weeks of the 2024–2025 season. As of November 1st of 2024, a total of 382 games had been played across these leagues. We first computed the actual SS(H) values using the Success Score formula, then compared them to the model's pre-match predictions to evaluate accuracy. This evaluation simulates a real-world deployment scenario, where accurate pre-match predictions can support decision-making in tactical planning, betting analysis, and performance evaluation. Overall, the model exhibited robust performance in predicting SS(H), achieving a Mean Squared Error (MSE) of 1.693 and a Mean Absolute Error (MAE) of 1.061 across all leagues. Table 4 presents the league-specific performance metrics. The English Premier League (ENG) achieved the lowest MAE (0.959), indicating the highest prediction accuracy for SS(H) in this league. Conversely, La Liga (ESP) recorded the highest MSE (1.862), reflecting greater variability in prediction accuracy for matches in this league. Ligue 1 (FRA) was not included in training; thus its results represent out-of-domain generalization for the model.
These results highlight the model's capacity to generalize from historical data to new match scenarios. Additionally, discrepancies between actual and predicted Success Scores across leagues offer valuable insight into context-specific factors affecting predictive accuracy, paving the way for future refinement.

Table 4. Predictive performance on unseen 2024–2025 matches (weeks 2–11), by league.
The model was trained on matches from 2020–2021 to 2022–2023. Metrics are Mean Squared Error (MSE) and Mean Absolute Error (MAE) for home Success Score predictions (lower is better).

League	MSE	MAE
ENG-Premier League	1.553	0.959
ESP-La Liga	1.862	1.107
FRA-Ligue 1	1.876	1.160
GER-Bundesliga	1.578	1.054
ITA-Serie A	1.561	1.030

Applications

We introduce two main applications for our parameter, the success score. The first application is evaluating team performance by comparing the predicted Success Score, which reflects a team's potential, with the actual Success Score achieved during games. This comparison helps determine whether a team performed above, below, or at
their expected potential, offering a detailed and objective evaluation of performance. The second application is predicting match outcomes by analysing the relationship between Success Scores and game results. By identifying specific thresholds within the Success Score range that correlate with outcomes such as wins, losses, and draws, the model enables effective classification and prediction of match results. This provides a reliable and structured approach for understanding and forecasting football match outcomes.

A. Team Performance Analysis Using Success Score

In this section, we analyse the performances of two teams with contrasting outcomes in the 2024–2025 season: FC Barcelona and Manchester United. These teams were selected to highlight distinct scenarios in team performance and illustrate how the Success Score can provide deeper evaluative insights. Manchester United faced significant challenges throughout the season, resulting in the dismissal of their head coach, Erik ten Hag, after nine weeks. In contrast, FC Barcelona emerged as one of the most consistent and high-performing teams, reaching the top of La Liga by week 11 and receiving widespread critical acclaim. By comparing the actual Success Scores achieved by these teams with the predictions generated by our model, we evaluated their performance across two categories:

I. Solid performance: This category includes matches in which a team met or exceeded its predicted potential. It is further divided into: *Exceeding Expectations:* The actual Success Score is higher than the predicted Success Score. *Dominant Performance:* Both actual and predicted Success Scores exceed 5, indicating control and superiority over the opponent. *As Expected in a Weaker Matchup:* Both actual and predicted Success Scores are below 5, reflecting that the team met expected performance in a difficult fixture.

II. Poor performance: Matches where a team underperformed relative to its predicted potential. Specifically, this occurs when: *Failure to Meet Expectations:* The predicted Success Score is above 5, but the actual Success Score falls below 5 indicating a failure to capitalize on predicted advantage.

By categorizing matches in this manner, we can effectively assess how well teams like FC Barcelona and Manchester United translated their predicted potential into actual results during the 2024–2025 season. This analysis provides insights into team consistency and episodes of underperformance. **Supplementary Table S1** presents the predicted and actual Success Scores for both teams, alongside match outcomes and the assigned performance categories.

Manchester United*: A Season of Missed Expectations*

The 2024–2025 season proved to be a challenging one for Manchester United, who began the campaign with high hopes under head coach Erik ten Hag. Despite a talented squad and strong predicted Success Scores, the team frequently failed to meet expectations in crucial matches. This pattern of underperformance ultimately led to a leadership change. Analysing matches from weeks 2 to 9 reveals several instances of this struggle. In Week 2, for example, United faced Brighton with a predicted advantage but lost 1–2, resulting in a "poor" performance classification. Such outcomes highlight their recurring inability to deliver on potential, especially in home games. Tactical rigidity and a lack of cohesion were also evident. A significant proportion of their matches during this period were classified as "poor" performances, underscoring consistent underachievement. This ultimately contributed to Ten Hag's dismissal, reflecting the club's urgent need to reorient its competitive strategy.

Barcelona*: A Consistently High-Performing Season*

Under the guidance of new manager Hansi Flick, Barcelona has demonstrated remarkable consistency and tactical efficiency. Despite an early setback in a 4–2 loss to Osasuna, the team showed strong form from weeks 2 to 11, often exceeding its predicted potential. One standout example occurred in Week 4 against Real Valladolid, where Barcelona not only fulfilled its predicted dominance but exceeded it with a commanding 7–0 win. Similarly, in Week 6, they defeated Villarreal 5–1 away from home, again outperforming their predicted Success Score. The majority of Barcelona's matches were classified as "solid" performances, indicating a strong alignment between potential and execution. Their consistent ability to meet or exceed expectations highlights their strategic adaptability and superior match-day execution under Flick's leadership, driving their ascent to the top of the La Liga table.

B. Utilizing the Relation Between Success Score and Football Match Outcomes

In this section, we aim to explore the relationship between the Success Score and football match outcomes. To achieve this, thresholds of the actual Success Score were identified that exhibit the highest correlation with match outcomes. To determine these thresholds, data from all matches during the 2020–2022–2023 seasons across four major football leagues was analysed. Threshold selection. Using matches from 2020–2021 to 2022–2023, we swept candidate cut-points for SS(H) and selected $\tau_low = 4.6$ and $\tau_high = 5.3$ to maximize out-of-fold macro-accuracy for the three-class mapping (Lose/Draw/Win). These fixed thresholds were then applied, without refitting, to 2024–2025 for evaluation. and are summarized in Table 5. This table presents the thresholds for the Success Score alongside the corresponding match outcomes. For example, 85.44% of teams with a Success Score in the range of 0–4.6 lost their matches, 13.46% ended in a draw, and 1.10% won. Similar trends were observed for other ranges. Overall, using these thresholds, 77% of the matches from those seasons ended with an actual outcome that matched their classified range.

Table 5. Outcome rates by actual Success Score threshold range (2020–2021 to 2022–2023).
Percentage of matches resulting in Loss, Draw or Win for three ranges of the actual SS(H): 0–4.6, 4.6–5.3 and 5.3–10. Row percentages sum to ~100%. SS(H), home Success Score.

Threshold Range	Loss (%)	Draw (%)	Win (%)
0–4.6 (Lose)	85.44	13.46	1.10
4.6–5.3 (Draw)	29.57	49.13	21.30
5.3–10 (Win)	1.10	16.48	82.42

C. Predicting Match Outcomes Using the Predicted Success Score

In this section, we predict match outcomes using the predicted Success Score as the basis for classification. To evaluate the model's performance, match data from the 2024–2025 season was analysed, covering weeks 2 to 11 across the top five European football leagues: the English Premier League (ENG), Ligue 1 (FRA), Bundesliga (GER), La Liga (SPA), and Serie A (ITA). The thresholds 5.3 and 4.6, identified in the previous section, were applied to classify match outcomes. These thresholds define two key binary classification scenarios: "Win" vs. "Not Win" and "Lose" vs. "Not Lose." We define the following binary classification scenarios:
Win vs. Not Win: A "Win" indicates the team won the game, while "Not Win" includes both draws and losses. Scores between 5.3 and 10 were classified as "Win," and scores below 5.3 as "Not Win." Lose vs. Not Lose: A "Lose" indicates the team lost, with a Success Score between 0 and 4.6, while "Not Lose" includes wins and draws.

The model demonstrated strong performance in both scenarios, as summarized in **Error! Reference source not found.**. For "Win vs. Not Win," the model correctly classified 280 out of 382 matches, achieving an accuracy of 73.30%. For "Lose vs. Not Lose," 287 out of 382 matches were correctly classified, with an accuracy of 75.13%. Performance varied by league, as shown in **Error! Reference source not found.**. The highest accuracy for "Lose vs. Not Lose" predictions were in Serie A (78.21%), while Ligue 1 achieved the highest accuracy for "Win vs. Not Win" predictions (75.00%). The Bundesliga had the lowest accuracy for "Lose vs. Not Lose" predictions (68.85%). The achieved accuracy rates of 73.30% for "Win vs. Not Win" and 75.13% for "Lose vs. Not Lose" indicate strong predictive performance, especially in the context of football match outcomes, which are inherently uncertain and influenced by numerous factors such as player performance, injuries, and in-game events. Football outcomes are widely recognized as challenging to predict due to their dynamic nature and the potential for upsets. Achieving accuracies above 70% suggests that the model successfully captures key patterns and trends in the data, providing a reliable method for classification. Furthermore, the accuracy compares favourably to other predictive approaches in football analytics, where models often struggle to exceed 60–70% due to the variability of match results. By leveraging the Success Score and its optimal thresholds, this model demonstrates an ability to consistently translate predictive metrics into actionable classifications, making it valuable for applications such as betting, performance analysis, and strategic decision-making.

Table 6. Outcome-classification accuracy on 2024–2025 matches (weeks 2–11), all leagues combined.
Accuracy for two binary tasks using the predicted SS(H): "Win vs. Not Win" (threshold ≥ 5.3 → Win) and "Lose vs. Not Lose" (threshold ≤ 4.6 → Lose). SS(H), home Success Score.

Metric	Win - Not Win	Lose - Not Lose
Correct Predictions / Total Predictions	280 / 382	287 / 382
Accuracy (%)	73.30%	75.13%

Table 7. Outcome-classification accuracy by league (2024–2025, weeks 2–11).
Per-league accuracy for “Lose vs. Not Lose” and “Win vs. Not Win,” using the SS(H) thresholds defined in Table 6. Leagues: ENG, ESP, FRA, GER, ITA. SS(H), home Success Score.

League	Lose - Not Lose Accuracy (%)	Win - Not Win Accuracy (%)
ENG-Premier League	76.5	72.50
ESP-La Liga	74.75	72.73
FRA-Ligue 1	76.56	75.00
GER-Bundesliga	68.85	73.77
ITA-Serie A	78.21	73.08

Discussion

The results of this study highlight the utility of the Success Score as a robust and interpretable metric for evaluating team performance and predicting football match outcomes. By integrating actual goals scored with advanced metrics such as Expected Goals, the Success Score provides a more holistic view of a team’s effectiveness on the pitch. The model demonstrated strong predictive accuracy, effectively capturing the nuanced dynamics of football matches and generalizing well to unseen data from the 2024–2025 season. However, league-specific variations—such as higher error rates in La Liga compared to the English Premier League—suggest the influence of contextual factors, including tactical preferences, fan behaviour, and competitive balance, offering directions for further model refinement. The dual applications of the Success Score—team performance evaluation and match outcome prediction—underscore its practical value. By comparing predicted and actual scores, the metric enables analysts to identify overperformance, underperformance, and consistency. This was exemplified in the contrasting 2024–2025 trajectories of FC Barcelona and Manchester United, where Barcelona consistently aligned with model expectations, while United frequently fell short. Such analysis can support coaching evaluations, performance diagnostics, and mid-season adjustments. For match outcome classification, the model achieved 73.30% accuracy for "Win vs. Not Win" and 75.13% for "Lose vs. Not Lose," surpassing many benchmarks in football analytics. Given the sport’s inherent unpredictability—driven by injuries, tactical adjustments, and in-game randomness—this level of performance highlights the model’s ability to distil complex match dynamics into actionable insights. The use of thresholds derived from Success Score distributions add interpretability and simplify real-world applications, including match preparation, broadcast analysis, and betting strategies. Our analysis of home Success Scores reinforced the significance of contextual factors. The observed home-ground advantage, especially in La Liga, suggests that incorporating features related to venue, crowd intensity, and travel distance may further enhance model performance. Future iterations could benefit from explicitly modelling these environmental factors.

Despite promising results, several limitations must be acknowledged. First, while the dataset spans multiple seasons and leagues, it may not capture long-term tactical evolutions or rare scenarios. Including more seasons, leagues outside Europe’s top tier, and international competitions could improve generalizability. Moreover, while match-level data is comprehensive, it does not account for real-time events such as injuries, red cards, or tactical substitutions, factors that often alter a game’s trajectory. Future models could incorporate live data streams to support in-game forecasting and decision support. Another avenue for refinement involves model interpretability. While neural networks are powerful, they are often viewed as black boxes. Employing explainable AI techniques such as SHAP values or permutation importance could offer insight into which features most influence predictions, helping analysts and coaches trust and act upon the model’s outputs. Similarly, visual tools like confusion matrices could help diagnose model strengths and limitations in different match contexts. Finally, while fixed thresholds were effective for outcome classification, they may need adaptive tuning across seasons or leagues to maintain optimal accuracy. Dynamic thresholding techniques or probabilistic classification frameworks could increase robustness in changing competitive environments.

In summary, the Success Score provides a scalable, interpretable, and effective framework for quantifying team performance and forecasting outcomes. Its applications span strategic planning, performance tracking, and predicting.

Conclusion

This study introduced a predictive framework centered on the Success Score; a novel performance metric that blends goals scored with Expected Goals to offer a more comprehensive evaluation of team effectiveness. Using a deep neural network architecture trained on three seasons of match data from Europe’s top football leagues, the model achieved strong predictive

accuracy and demonstrated generalizability to unseen data from the 2024–2025 season. The model's strength lies in its feature-rich design, combining tactical play style indicators, rolling performance averages, and team-level quality and prestige attributes. This multi-dimensional feature set enabled the model to capture both short-term form and longer-term consistency, contributing to its robust forecasting performance.

The Success Score proved useful in two key applications: assessing whether teams met or missed their expected potential and predicting match outcomes through interpretable classification thresholds. These insights have value for analysts, coaches, and performance staff seeking to evaluate strategies, monitor trends, or prepare for opponents.

Looking ahead, future work should focus on extending the dataset to include additional leagues and competitions, incorporating real-time data streams, and enriching the model with player-level and in-game dynamics. Adding interpretability tools could also strengthen adoption in applied settings by making predictions more transparent and actionable.

With continued refinement, the Success Score and its predictive framework offer a powerful, scalable approach to advancing data-driven decision-making in modern football.

Data Availability Statement

All notebooks are available at: GitHub. https://github.com/farzammnf/Success-Score

FBREF's Dataset available at: FBREF Data. https://github.com/farzammnf/Success-Score/tree/main/Data/Fbref

SoFIFA's Dataset available at: SOFIFA Data. https://github.com/farzammnf/Success-Score/tree/main/Data/Sofifa

Data used for the Model: Dataset. https://github.com/farzammnf/Success-Score/tree/main/Data/Model%27s%20data

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Author Contributions

F.M. (Farzam Manafzadeh) conceived the study, developed the methodology, implemented the analyses, and drafted the manuscript. C.E. (Changiz Eslahchi) co-conceived the study, contributed to methodology and analysis verification, supervised the research, and revised the manuscript. H.N. (Hadi Nobari) supervised the research, provided guidance, and contributed to manuscript revision. All authors reviewed and approved the final manuscript and agree to be accountable for all aspects of the work. H.N. is the corresponding author (Hadi.Nobari1@gmail.com).

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Competing interests

The author(s) declare no competing interests.

Ethics statement

This study investigates publicly available performance data and involved no human participants, personal data, or intervention. In accordance with the policies of the authors' institution, formal ethics approval and informed consent were not required for this research.

Supplementary Files

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